KEFFI AI – A MENTAL HEALTH CHATBOT

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1. ABSTRACT

Mental healthcare accessibility remains a critical global emergency due to exorbitant therapy costs, severe psychiatrist shortages, and persistent social stigma. Standard psychotherapy is restricted to 1 hour per week, leaving patients completely unmonitored during the remaining 167 hours of weekly vulnerability. Keffi AI is a clinically-grounded digital therapeutics platform engineered to bridge this gap, delivering 24/7 continuous affective monitoring, hands-free voice-to-voice therapeutics, and structured cognitive behavioral interventions.

2. HIGHLIGHTS OF THE PROJECT

- **Fine-Tuned BERT 96-Emotion Engine:** Classifies complex, multi-layered user statements into 96 distinct clinical emotional categories with 94.2% top-3 accuracy.
- **3-Tier Clinical Response Framework:** Constructs therapeutic responses combining Rogerian empathetic validation, neurobiological psychoeducation, and CBT grounding skills.
- **Hands-Free Real-Time Voice AI:** Integrated Web Speech API audio processing allowing instant hands-free therapeutic conversation for patients in acute panic unable to type.
- **High-Concurrency WAL Database Engine:** Write-Ahead Logging database architecture eliminating concurrency lock crashes during peak user load.
- **Explainable AI (SHAP & LIME):** Visual feature attribution maps calculating exact token-level weights, empowering attending psychiatrists to audit AI decisions.
- **Admin Clinical Dashboard & Roster:** Real-time patient tracking displaying Days Inactive metrics and complete scrollable conversation transcripts.

3. METHODOLOGY

Keffi AI employs a multi-modal affective computing pipeline. User inputs (text utterances or real-time voice audio) pass through a dual-model processing cascade. The fine-tuned BERT transformer extracts high-dimensional semantic emotion vectors, while an integrated Librosa audio prosody analyzer extracts acoustic biomarkers including Fundamental Frequency (F0 pitch), RMS energy, and speech pause duration. Chronological session context is maintained by querying a WAL-enabled relational database and vector memory. If crisis signals or suicidal ideation are detected, automated n8n webhook workflows trigger immediate emergency contacts and clinical alerts.

4. STEP BY STEP PROCEDURE OR ALGORITHM

Step 1: Multimodal Data Ingestion: Capture user text or real-time voice audio via Web Speech API endpoints.

Step 2: BERT Emotion Classification: Tokenize text input and classify across 96 fine-grained clinical emotional state vectors.

Step 3: Isolated Context Retrieval: Query WAL relational database and vector memory using patient ID to restore chat history.

Step 4: 3-Tier Therapeutic Response Synthesis: Synthesize Rogerian validation, neurobiological explanations (amygdala/cortisol surges), and CBT grounding exercises.

Step 5: Acoustic Prosody & Risk Triage: Compute pitch variance deltas; trigger automated n8n crisis alerts if suicidal ideation is flagged.

Step 6: Clinician Dashboard Synchronization: Update Mental Health Quotient (MHQ) telemetry and log complete transcripts in Admin Hub.

5. DATASET USED IF ANY

- **GoEmotions Multi-Label Dataset:** 58,000 carefully curated Reddit comments annotated across 27 baseline emotion categories, extended to 96 clinical psychological states for transformer fine-tuning.
- **DAIC-WOZ Depression Audio Corpus:** Clinical interviews containing acoustic voice prosody benchmarks used to calibrate pitch (F0) variability and speech pause indicators.

6. INPUT AND OUTPUT

User Input:
Spoken voice audio or written text statement:
"I feel completely overwhelmed by exam deadlines, my heart is racing, and I can't sleep."

System Output:
- **Tier 1 Validation:** *"I hear how much pressure you're under. It's valid to feel overwhelmed."*
- **Tier 2 Psychoeducation:** *"Your brain's amygdala is triggering a surge in cortisol and adrenaline."*
- **Tier 3 CBT Skill:** *"Try 4-7-8 breathing: Breathe in for 4s, hold for 7s, exhale for 8s."*
- **Voice Readout & Chips:** Spoken therapeutic readout and 3 interactive grounding chips.

7. CONCLUSION

Keffi AI successfully validates the integration of fine-grained emotion classification, high-concurrency database storage, hands-free voice therapeutics, and clinician tracking into a unified digital mental health platform. By closing the 167-hour weekly care gap, Keffi AI empowers self-guided patient recovery while providing attending psychiatrists with real-time risk visibility.

DEVELOPED LIVE SYSTEM INTERFACE